TEZPUR UNIVERSITY

Roof Truss Design

GROUP-E

STRUCTURAL DESIGN-II (CE328)

Mr. Debaraj Bailung Sonowal

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This report has following contains: -

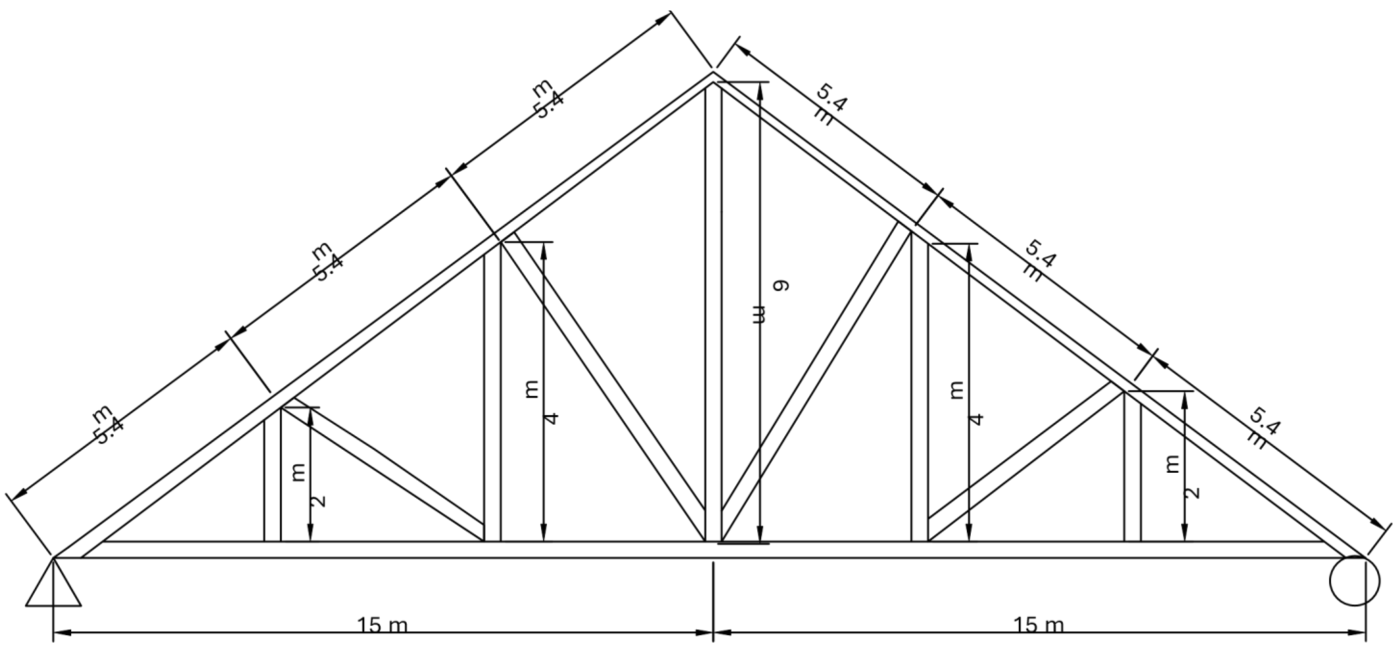
1.Diagram of truss

2.Load calculation for truss (Manually)

3.Analysis of Truss by using SAP2000

4.Design of member of truss

1.Diagram of truss



RESIDENTAL BUILDING TRUSS

2.Load calculation for truss (Manually)

1. Dead Load:

Weight of sheeting including laps and connection

$= 170\ N/m^2$

Weight of purlins $= 120\ N/m^2$

Self weight of truss $= 20 + 6.6L$

$= 20 + 6.6 \times 30$

$= 218\ N/m^2$

Total dead load $= 170 + 120 + 218$

$= 508\ N/m^2$

Each purlin take care an area $= 5.4 \times 5$

$= 27\ m^2$

Total dead load $= 27 \times 508$

$= 13.716\ kN$

$= 13.71\ kN$

2. Live Load:

Span of truss = 30 m
Pitch = 1/5 , Rise = (1/5 × 30) = 6 m

$\tan \Theta = 6 / 15$
$\Theta = 21.80°$

Live load on purlin
= 750 – (21.8 – 10) × 20
= 514 N/m² (from Cl 4.1 IS875 Part 2)

Spacing of purlins = 5.4 m

Live load = 514 × 5.4
= 2775.6 N/m

Concentrated live load = 2775.6 × 5
= 3.4 kN

3. Wind Load:

$V_z = V_b \times K_1 \times K_2 \times K_3 \times K_4$ (from Cl 6.3 IS875 Part 2)

Where,

$V_b \rightarrow$ (from Cl 6.2 IS875 Part 2)

$K_1 \rightarrow$ (from Cl 7.1 IS875 Part 2)

$K_2 \rightarrow$ (from Cl 7.2 IS875 Part 2)

$K_3 \rightarrow$ (from Cl 6.3.3 IS875 Part 2)

$K_4 \rightarrow$ (from Cl 6.3.4 IS875 Part 2)

$Vz = 50 \times 1 \times 1.05 \times 1 \times 1$

$Vz = 52.50$ m/s

(from Cl 7.2 IS875 Part 2)

Wind pressure at 'Z' elevation (Pz) $= 0.6 \times Vz^2$

$= 0.6 \times (52.5)^2$

$= 1653.75$ N/m² (from Cl 7.2 IS875 Part 2)

Design wind pressure (P_d) = $K_d \times K_a \times K_c \times P_z$

Where,

K_d → (from Cl 7.2.1 IS875 Part 2)

K_a → (from Cl 7.2.2 IS875 Part 2)

K_c → (from Cl 7.3.3 IS875 Part 2)

$$P_d = 0.9 \times 1 \times 0.9 \times 1653.75$$
$$= 1071.63 \text{ N/m}^2$$

Tributary area of truss = $2 \times 5.4 \times 15$

$= 162 \text{ m}^2$

Net wind load = (Cpe – Cpi) × A × Pd

Where,

Cpi = +0.7 , -0.7 (from Cl 7.3.2.2 IS875 Part 2)

Cpe = h/w = 15/30 = 3/6 = ½

When wind angle is 0° (from Cl 6.3 IS875 Part 2)

On windward side (C_{pe}) = $-0.4 + (1.8 / 10) \times 0.4$

$= -0.328$

On leeward side (C_{pe}) = -0.4

When wind at 90°; for rafter slope = 21.8°

On windward side (C_{pe}) = -0.7

On leeward side (C_{pe}) = -0.6

Force (F) = $(C_{pe} - C_{pi}) \times A \times P_d$

$= (-0.7 - 0.7) \times 162 \times 1071.63$

$= -243045.68$

$= +243.045$ kN (Windward)

Leeward:

$F = (Cpe - Cpi) \times A \times Pd$

$= (-0.6 - 0.7) \times 162 \times 1071.63$

$= -225685$ N

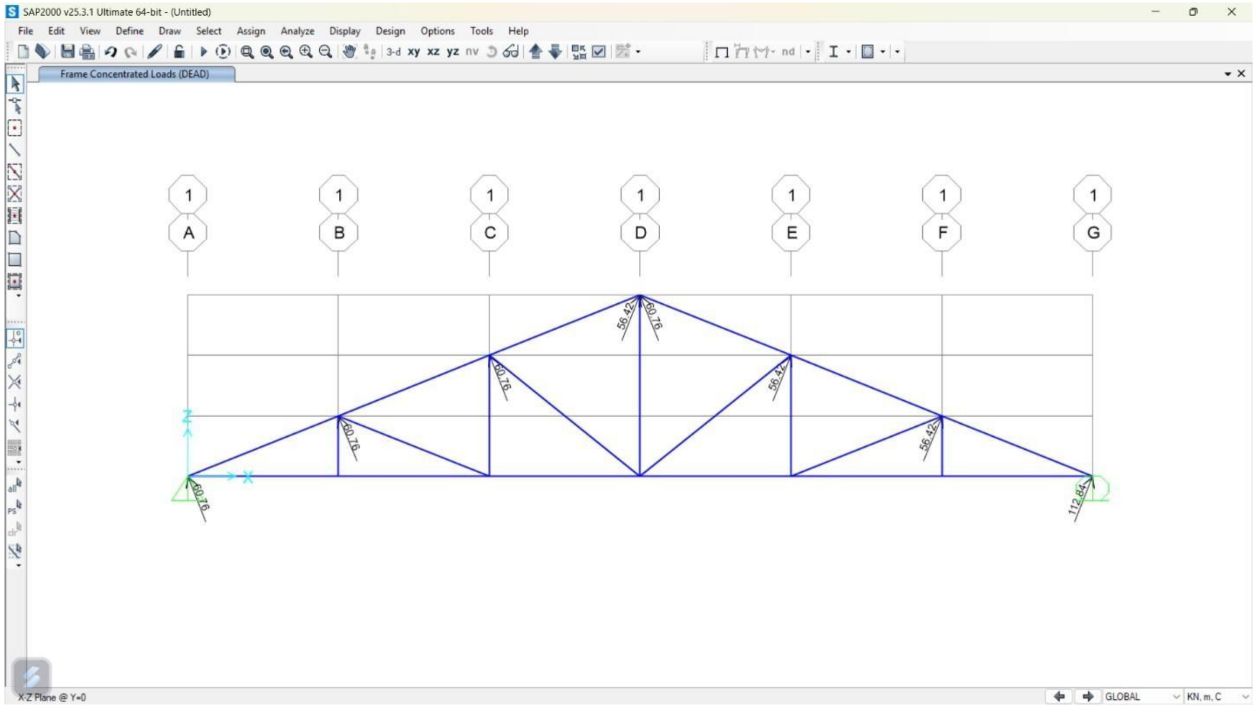
$= +225.685$ kN

Total dead load + Total live load

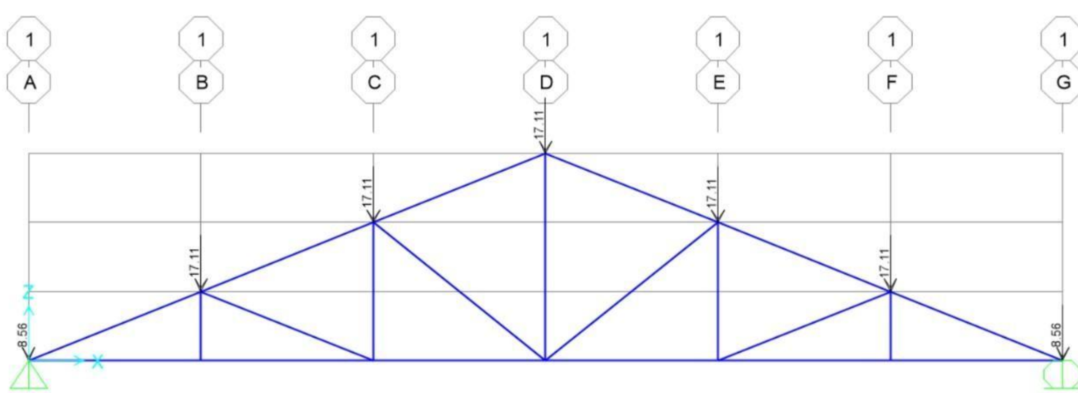
= 13.71 + 3.4

= 17.11 kN

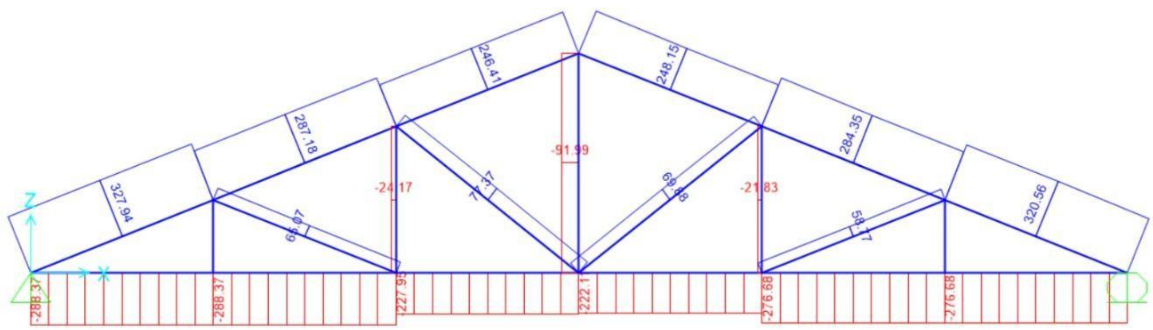
3. Analysis of truss by using SAP 2000 :



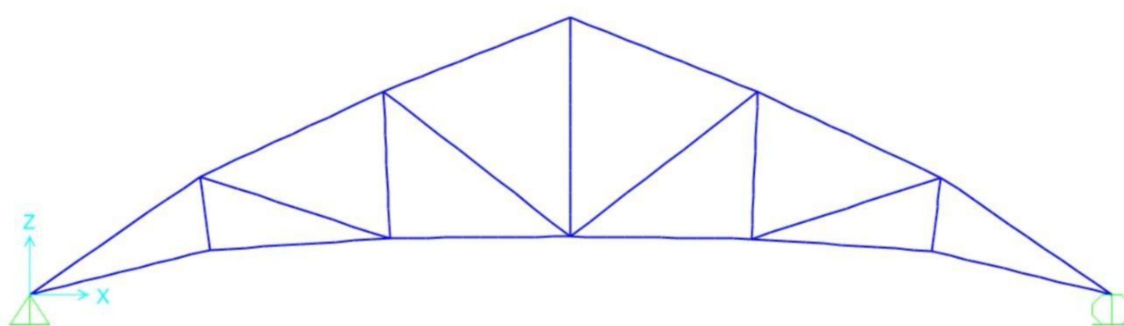
Wind load at truss



Dead load at truss



Axial load on truss (max load is 288.37 KN)



Deformed shape Joint load table:-

TABLE: Joint Reactions

Joint	OutputCase	CaseType	F1	F2	F3	M1	M2	M3
Text	Text	Text	KN	KN	KN	KN-m	KN-m	KN-m
1	DEAD	LinStatic	6.447	0	-169.653	0	0	0
7	DEAD	LinStatic	0	0	-162.883	0	0	0

4. Design of member of truss :

Maximum compressive force = 288.37 KN

$T_{dg} = (A_g \times f_y) / \gamma_{mo}$

$288.37 \times 10^3 = (A_g \times 250) / 1.1$

$A_g = (288.37 \times 10^3 \times 1.1) / 250$

$A_g = 1268.82 \text{ mm}^2$

We are choosing ISA9060

Area of cross section = 1401 mm²

Thickness = 10 mm